2013 American Control Conference (ACC)
Washington, DC, USA, June 17-19, 2013

A real-time control system architecture for industrial power amplifiers

Faran A. Qureshi *Student Member, IEEE*, Veaceslav Spinu *Student Member, IEEE*, Korneel Wijnands, Mircea Lazar *Member, IEEE*

***Abstract*— Power amplifiers are a highly important component in a range of industrial applications, such as, servo-drives, magnetic resonance imaging, energy systems, and audio. The control system for power amplifiers should satisfy a range of requirements, e.g., offset free tracking, stability margins, and fast transient response. The intrinsic switching behaviour of modern power amplifiers and the sampled nature of digital control systems represent additional design challenges. This paper presents a complete development cycle for a cascaded digital control system for an industrial current amplifier. The inner control loop contains an optimal state-feedback controller and observer to ensure fast transient response. To guarantee reference-to-output frequency-domain specifications an outer control-loop is designed. Lastly, a rate-limiter is added to prevent clamping in the control input and overstress of the amplifier components. A special attention is given to synchronization in between sub-assemblies of the control system within the FPGA implementation. The paper concludes with the real-time measurements and comparison with the original control systems of the industrial amplifier.**

***Index Terms*— Real-time control, High precision, Industrial current amplifier, FPGA.**

I. INTRODUCTION

High precision current amplifiers have become a key component of motion systems, health care industry, electronic optic systems, and many other applications [1], [2]. All these applications show continuous growth in performance, which imposes tight specification on each of their components. Meeting these specifications with conventional control becomes harder and harder, thus motivating further research in design and implementation of control systems tailored to *power amplifiers* (PA).

There are several reasons why development of a control system for a PA is a challenging task. Firstly, there are multiple indices which define the performance of the PA, e.g., step response time, overshoot, sensitivity to disturbance, bandwidth, and phase delay. These indices are very different in nature and present a mix of time-domain and frequency-domain specifications. Secondly, modern PA are switching devices with fast dynamics, subject to hard constraint on control inputs and system states. And lastly, the available measurements are also subject to delays.

The existing practice in industry is to design a loop shaping controller to achieve frequency-domain specifications from reference to output [3]. Often there is a clear time-scale separation in between the dynamics of the load and the output filter of the amplifier. Concentrating only on the dominant dynamics, i.e., load dynamics, may result in undesired oscillations in the output filter during transients. The conventional solution to reduce these oscillations is to wrap the power stage in a damping loop first, see, e.g., [4]. Although, this solution stabilizes the oscillations, the additional damping may have adverse effects on the transient performance and impedance of the PA.

In recent years, the *model predictive control* (MPC) emerged as a serious contender in control of power converters [5], [6], as it handles well constraints and optimizes the transient behaviour of the system. However, there are several major disadvantages of the MPC in the high precision industrial applications. It is difficult to design an MPC controller which would respect given frequency-domain specifications [7]. The sampling rates of precision PA are in hundreds of kilohertz range, which makes online MPC inapplicable. For large systems with stiff dynamics, such as, H-bridge PA with filtering and large inductive load, the number of regions in the explicit MPC [8] soars beyond several thousands, thus, making this approach also infeasible for practical applications. Due to these reasons, a real-time MPC controller is not yet suitable for fast and high precision industrial amplifiers.

Apart from MPC several state-feedback solutions where proposed for control of power amplifiers [9]–[11], and showed improved PA performance in laboratory experiments. However, the industrial acceptance of this technology is still very limited. In-line with these observations, the aim of this work is to propose a design flow and a real-time digital control architecture for an industrial current amplifier, overcoming all the real world challenges. These challenges include, measurement synchronization, ensuring fast transient response, guaranteeing frequency-domain specifications for stationary operation, constraints satisfaction, and FPGA implementation. It is proposed to design the *output feedback controller* (OFC) to comply with the frequency domain specifications, such as, offset-free tracking and high impedance of the amplifier at low frequencies. To ensure fast transient response optimal *linear-quadratic regulator* (LQR) is employed together with a Luenberger observer [12]. In contrast to typical implementation of the state-feedback in power electronics, e.g. [10], where the feedback gain multiplies the state of the system directly, in this paper the feedback is applied to the error in between the current state of the system and the reference induced steady-state. Faster step response

F. A. Qureshi is with the Laboratoire d'Automatique, École Polytechnique Fédérale de Lausanne, Switzerland. V. Spinu, K. Wijnands and M. Lazar are with the Department of Electrical Engineering, Eindhoven University of Technology, The Netherlands. E-mails: faran.qureshi@epfl.ch, {v.spinu,c.g.e.wijnands,m.lazar}@tue.nl.

978-1-4799-0178-4/$31.00 ©2013 AACC

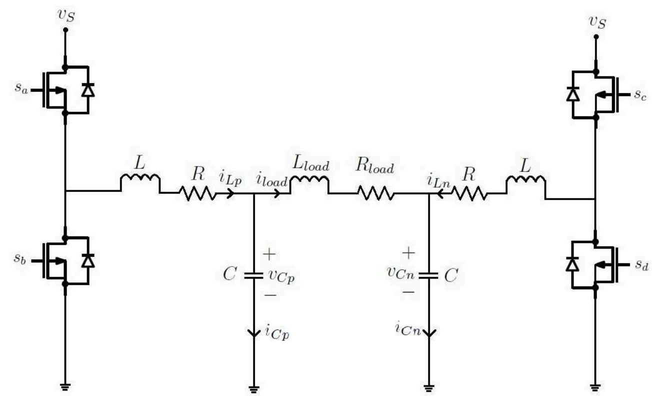


Fig. 1: The schematics of the current amplifier.

is achieved in this way. The last ingredient of the control architecture is the rate of change limiter of the reference. This is a simplest implementation of a reference governor [13]. It reduces the clamping of the control input and enforces constraints satisfaction on the system states. The proposed control architecture was implemented in an FPGA on-board of an industrial current PA without any modification to the original power-stage design.

II. Problem Formulation

The schematic of the considered high precision industrial current amplifier is depicted in Fig. 1. Therein, L denotes the inductance of the coil, C the capacitance, R the parasitic resistance and v_s the DC source voltage, while R_{load} and L_{load} are the load resistance and inductance, respectively. The current through the inductors and the voltage across the capacitors is represented by i_L and v_C with subscripts p and n, which denote the positive and negative leg of the amplifier, respectively. The available measurements are the load current i_{load} and the currents through the two filter capacitors i_{Cp} and i_{Cn}; see Fig. 1. The $i_{load},\ i_{Cp}$ and i_{Cn} are measured synchronously. However, the measurements are available for control action computation with different delay. The power-stage is controlled by two PWM signals, which are updated twice per cycle with the frequency f_s. First PWM waveform drives the switches S_a and S_b, and the other waveform drives the switches S_c and S_d. There is dead-time introduced to prevent cross-conduction through the switches of the same leg. The parameters of the power stage are summarized in Table I. This work is to achieve following three objectives:

Objective 1 *Design a real-time control-system architecture for the implementation of digital controllers in an industrial current amplifier.*

Objective 2 *Attain fast tracking of the digital set-point of the load current i_{load} with good transient response (overshoot, settling time etc.).*

Objective 3 *Achieve frequency domain and stationary performance specifications i.e. a bandwidth of at least $6kHz$ from the digital set-point to the output load current i_{load} and a maximum output sensitivity of $-80dB$ at low frequencies i.e., 1 Hz.*

TABLE I: Parameter Values

Parameter	Value
R_{load}	5.36Ω
L_{load}	3.48mH
R	0.492Ω
L	220μH
C	100nF
f_s	375kHz

Standard circuit modelling techniques [14] are employed to obtain the average continuous-time state-space model.

$$\dot{x}(t) = A_c x(t) + B_c u(t) \tag{1a}$$

$$y(t) = C_c x(t), \tag{1b}$$

where $A_c \in \mathbb{R}^{5\times5}$, $B_c \in \mathbb{R}^{5\times2}$ and $C_c \in \mathbb{R}^{3\times5}$. The system states are $x(t) = [i_{Lp}(t), v_{Cp}(t), i_{Ln}(t), v_{Cn}(t), i_{load}(t)]^\top \in \mathbb{R}^5$ and input is $u(t) = [d_1(t), d_2(t)]^\top$. $d_1(t)$ and $d_2(t)$ are the two duty cycles. The resulting matrices A_c, B_c and C_c are:

$$A_c = \begin{pmatrix} -\frac{R}{L} & -\frac{1}{L} & 0 & 0 & 0 \\ \frac{1}{C} & 0 & 0 & 0 & -\frac{1}{C} \\ 0 & 0 & -\frac{R}{L} & -\frac{1}{L} & 0 \\ 0 & 0 & \frac{1}{C} & 0 & \frac{1}{C} \\ 0 & \frac{1}{L_{load}} & 0 & -\frac{1}{L_{load}} & -\frac{R_{load}}{L_{load}} \end{pmatrix},$$

$$B_c = \begin{pmatrix} \frac{v_s}{L} & 0 \\ 0 & 0 \\ 0 & \frac{v_s}{L} \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad C_c = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}. \tag{2}$$

The state space model (1) is discretized with the zero-order-hold method and a sampling rate f_s, to obtain a discrete-time averaged state space model of the form:

$$x(k+1) = Ax(k) + Bu(k) \tag{3a}$$

$$y(k) = Cx(k) \tag{3b}$$

where $k \in \mathbb{Z}_+$, $A \in \mathbb{R}^{5\times5}$, $B \in \mathbb{R}^{5\times2}$, $C \in \mathbb{R}^{3\times5}$, $x(k) = [i_{Lp}(k), v_{Cp}(k), i_{Ln}(k), v_{Cn}(k), i_{load}(k)]^\top$ and $u(k) = [d_1(k), d_2(k)]^\top$.

III. Real-Time Control Architecture

The solution to Objective 1 is presented in this section. The main components of the proposed architecture along with all the signals coming in and going out of the digital-domain (FPGA) are depicted in Fig. 2. The digital reference signal is converted to the analog form by a DAC, and the reference, in analog domain, is subtracted from the measured load current i_{load} and is passed through a low-pass filter and a conditioning circuit, before it is discretized (using ADC) to obtain the signal e. Inside FPGA, the signal e is compensated for the analog conditioning circuitry, and the measurement delays. The load current i_{load} measurement is reconstructed inside the FPGA using the available reference signal and the measured error signal e. This approach to obtain the load current i_{load}, instead of directly discretizing the measured load current is used to achieve required noise specifications.

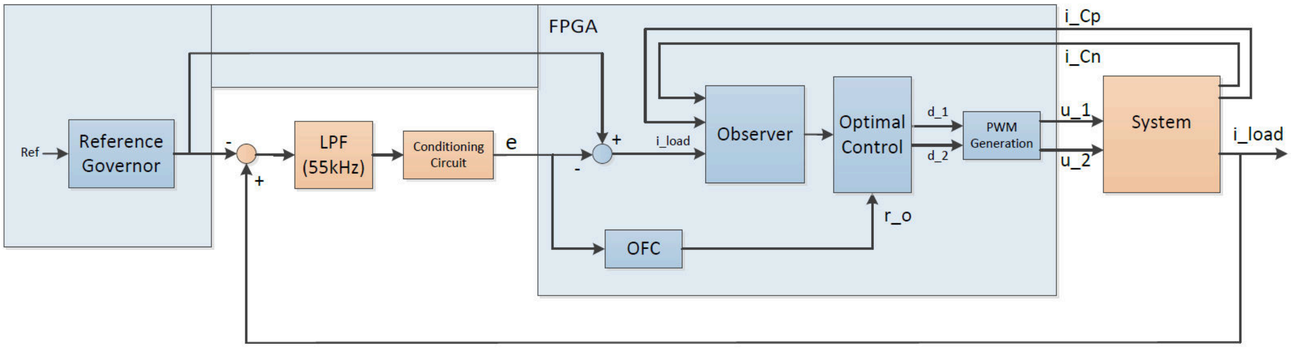


Fig. 2: Real-Time Control Architecture.

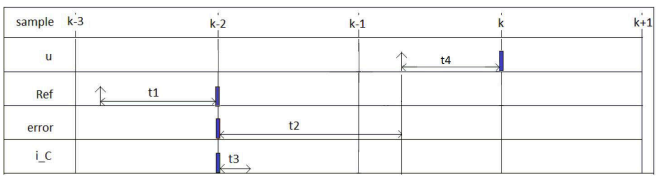


Fig. 3: Timing and synchronization diagram of the control-system. The illustration contains the time instants required for the computation of $u(k)$ (t_1*- total time required for sending the digital reference to the analog domain,* t_2*- measurement delay for the error signal e,* t_3*- measurement delay for the two capacitor currents,* i_{Cp} *and* i_{Cn}*,* t_4*- time available for calculations).*

Two other available measurements, i_{Cp} and i_{Cn} are also discretized using an ADC and are available within the FPGA. All other signals labelled in Fig. 2 are the same as introduced in Section II. The control inputs, u_1 and u_2 are the two PWM waveforms that drive the current amplifier. The control configuration includes a rate limiter to prevent saturation of the control input, an observer to estimate all the system states, an optimal controller for good transient response and an output feedback controller for achieving the stationary frequency-domain specifications. The detailed purpose and design of each of these individual components is explained in Section IV. The implementation of the proposed digital control scheme requires all the three measurements and the update of the control input to be synchronized. Additionally, synchronization of the measurements and the control inputs with the internal PWM carrier is also required to capture the averaged dynamics of the PA. Otherwise, the model from Section II may not be valid.

The synchronization of the control input application, measurements of the load current and capacitor currents is illustrated in Fig. 3. Sampling instants $k-3$ ... $k+1$ are scheduled with the sampling frequency f_s. The position of the sampling instants is slightly delayed with respect to the PWM carrier to compensate for delays in transistor drivers. Before the computation of $u(k)$ can begin, $ref(k-2)$, $ref(k)$, $e(k-2)$, $u(k-2)$, $u(k-1)$, $i_{Cp}(k-2)$, $i_{Cn}(k-2)$, $i_{Cp}(k-1)$ and $i_{Cn}(k-1)$ are required. The time instants at which all these signals become available, are shown in Fig. 3. Among the measurement signals, the output-error signal is available with the longest delay, due to higher precision required for this measurement. Hence, when the error signal $e(k-2)$ becomes available, all the other measurements are already available and the computation of the control action $u(k)$ may begin. Note that, for the simplicity of illustration, Fig. 3 shows the timings for the calculation of the control inputs at only one sample instant. The same steps are required at each sampling instant, and are implemented in a pipeline.

IV. Controller Design

With the control-system architecture defined in Section III, one can proceed with designing each individual subsystem. The main task within the Objective 2 is to ensure good tracking of the reference signal. Considering that the update rate of the digital set-point in this particular setup is limited to 10kHz and the sampling rate is of 375kHz one may conclude that the reference tracking problem is reduced to tracking of piecewise constant references. In this context, it is proposed to employ LQR to stabilize the amplifier at a specific constant output current. In contrast to classical solutions that rely on decoupling and SISO controllers, LQR enables the use of complete state-information and control freedom to optimally track the piecewise constant reference.

1) Linear Quadratic Regulator: Given the linear system

$$z(k+1) = Az(k) + Bs(k)$$

and the cost function

$$\mathcal{J} = \sum_{i=k}^{i=\infty} z(i)^\top Q z(i) + s(i)^\top R s(i), \tag{4}$$

the aim of the LQR is to find the control action $s(i)$ which minimizes $\mathcal{J}$. When there are no restrictions on the control action and system states the solution to LQR problem is available in the closed form $s(k) = Kz(k)$, where, $K = -(R + B^\top PB)^{-1}B^\top PA$ and P is the solution to the discrete-time Algebraic Riccati equation: $P = A^\top PA + Q - A^\top PB(R + B^\top PB^{-1})B^\top PA$. The control problem of a PA is to track the reference signal for the output current. Hence, at each sample time the LQR problem in translated coordinates must be solved, i.e., with $z = x - x_s$, and $s = u - u_s$ where x_s is the equilibrium state, such that $i_{load} = r_o$, and u_s is the corresponding control input.

Observe that the solution to the tracking problem becomes

$$u^*(x) = K(x - x_s) + u_s. \tag{5}$$

2) Steady States: At steady-state condition, the impedance of capacitors and inductors is infinite and zero, respectively. This assumption along with the requirement to have both duty-cycles $d_1 = d_2 = 0.5$, when i_{load} is zero, uniquely defines the relationship in between the reference r_o of the optimal controller, and the steady-state x_s and control input u_s, of the system, i.e.,

$$\begin{aligned} x_s &= Xr_o + f_1 \\ u_s &= Ur_o + f_2 \end{aligned} \tag{6}$$

where $X = \begin{bmatrix}1 & \frac{R_{load}}{2} & -1 & \frac{-R_{load}}{2} & 1\end{bmatrix}^T$, $f_1 = \begin{bmatrix}0 & \frac{v_s}{2} & 0 & \frac{v_s}{2} & 0\end{bmatrix}^T$, $U = \begin{bmatrix}\frac{R}{vs} + \frac{R_{load}}{2v_s} & -\frac{R}{vs} - \frac{R_{load}}{2v_s}\end{bmatrix}^T$, and $f_2 = \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\end{bmatrix}^T$.

3) Observer Design: Full state measurement is not available, and the measurements are subject to delays. To compensate for this delay and the time spent for computation of the control law, in this paper, it is proposed to employ a combination of a Luenberger observer [12]

$$\hat{x}(k-1) = A\hat{x} + Bu(k-2) + L(y(k-2) - C\hat{x}(k-2)), \tag{7}$$

with one step predictor

$$\hat{x}(k) = A\hat{x}(k-1) + Bu(k-1). \tag{8}$$

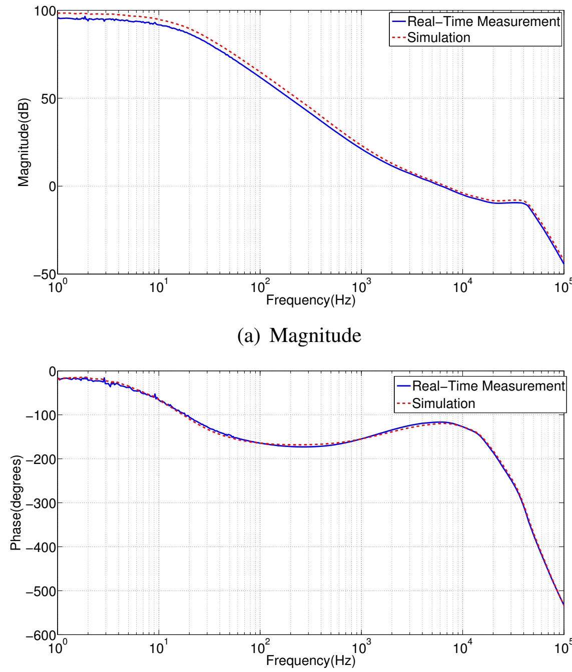
Since the disturbance in power electronics is not necessarily zero mean, e.g., additive dead-time effects, a high gain Luenberger observer is used instead of a Kalman filter. Based on the separation principle for linear systems, the correction matrix L can be designed separately from the LQR gain. Nevertheless, in this paper, L is designed such that the closed loop error dynamics, $\tilde{x}(k+1) = (A+BK-LC)\tilde{x}(k)$, where $\tilde{x} := x - \hat{x}$, are six times faster than the dynamics of the closed loop system $z(k+1) = (A+BK)z(k)$. This is to have sufficient time scale separation between the observer and the controller.

The computations involved in the optimal state-feedback control loop can be summarized in the following procedure:

- Compute x_s and u_s from (6).
- Compute observed states $\hat{x}(k-1)$ from (7).
- Compute the prediction $\hat{x}(k)$ from (8).
- Compute the current control action
  $u(k) = K(\hat{x}(k) - x_s) + u_s$.

A. Output Feedback Controller

The reference tracking optimal controller designed in previous subsection focuses on transient response and is expected to achieve Objective 2. However, it lacks integral action which results in steady-state offset in case of model / plant mismatch. This offset is highly undesirable in high-precision applications. Furthermore, the LQR design does not provide any information about output impedance of the PA. As these specifications are quantifiable in frequency-domain,



(a) Magnitude

(b) Phase

Fig. 4: Simulation and Real-Time measurement: Open-Loop Bode Plot.

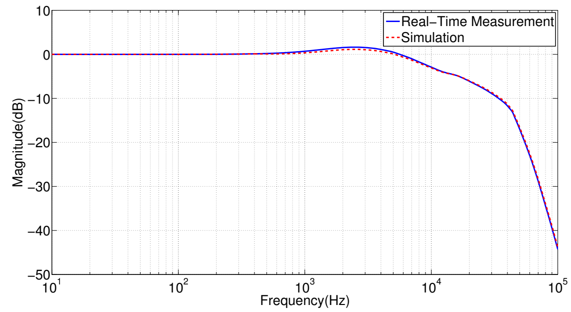
it is proposed to employ an outer control loop with a *single-input single-output* (SISO) controller designed to respect these specifications, and to achieve Objective 3. Based on the separation principle [15] for linear systems, the observer is not considered during the design of the OFC.

1) Plant model for the OFC: The plant for the design of the OFC includes only the closed-loop system with LQR feedback. However, in reality, the error signal which is the input of the OFC, is directly measured, and is subject to measurement delays and is also affected by a low-pass filter in its signal path, as depicted in Fig. 2. Hence, the plant consists of three components. First, the closed loop system from reference r_o to load current i_{load}. This is obtained by substituting the optimal control input (5), and the relation between the reference r_o and steady-states (x_s, u_s), (6), to the system model (3),

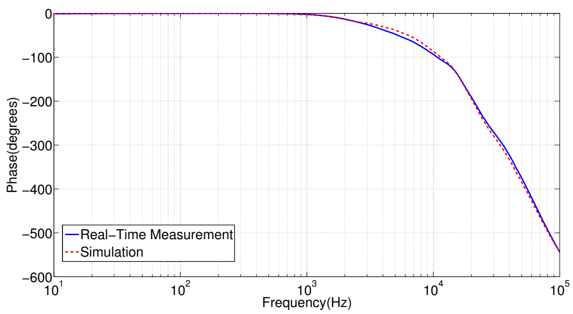
$$\begin{aligned} x(k+1) = (A+BK)x + (BU - BKX)r_o& \\ + Bf_2 - BKf_1& \end{aligned} \tag{9a}$$

$$y(k) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \end{pmatrix} x(k). \tag{9b}$$

Second, the measurement delay of two samples, which is modelled with the discrete-time transfer function: $delay(z) = \frac{1}{z^2}$. Third, a second-order low-pass filter with a cut-off frequency of 55kHz, and a continuous-time transfer function of the form: $Filter(s) = \frac{1}{\frac{1}{(2\pi f)^2}s^2 + \frac{1}{\sqrt{2}\pi f}s + 1}$, where $f = 55$kHz is the cut-off frequency. The SISO system (9), delay system, and the filter transfer function are combined to obtain the continuous-time model of the plant used for the OFC design. The OFC is designed using the Loop Shaping techniques [16]. The aim is to shape the frequency response of the open-loop transfer function, which includes the trans-



(a) Magnitude



(b) Phase

Fig. 5: Simulation and Real-Time measurement: Close-Loop Bode Plot.

fer function of the OFC and the plant. OFC consist of a gain, two poles and two zeros. The location of the poles, zeros and the gain of the controller are adjusted, in order to achieve performance specifications. The designed continuous-time transfer function of OFC is discretized and is converted to discrete-time state space form for implementation.

With the OFC, Objective 3, presented in Section II is achieved, and the steady-state error is also removed. However, the possible saturation of control inputs $u(k)$ and overstress of amplifier components, during transients, has not been addressed yet. This problem is solved by adding a rate limiter.

2) Rate Limiter: A rate limiter is added to the control configuration, as depicted in Fig. 2. The purpose of the rate limiter is to prevent saturation of control inputs $u(k)$ and limit the system states $x(k)$ within safe bounds during transients. It is tuned to limit the input reference to a sinusoidal of frequency 8kHz and amplitude 1A. The low frequency and steady-state behavior of the designed controller is unaffected while the transient response is improved. A less conservative, but more sophisticated solution to this problem is given by the *reference governor*. The interested reader is referred to [13], and references therein, for further details on this subject.

B. Simulations

To validate the control scheme, a simulation environment is created in SIMULINK, where the complete control architecture of Fig. 2 is implemented. Numerical values of design parameters for each subsystem are explained in what follows.

1) Optimal Controller: The weighting matrices Q and R, of the optimal controller's cost function (4) are designed to have good transient response. Maximum weight is applied to the state representing the load current i_{load}, followed by the two capacitor currents i_{Cp} and i_{Cn}. Following values of the weighting matrices were considered during design: $Q = diag[10, 0.05, 10, 0.05, 20]$, and $R = diag[10, 10]$.

2) OFC: The open-loop Bode plot for the OFC design is depicted in Fig. 4. As seen in the figure, the open-loop gain at low frequencies is designed to be around 100dB, and a bandwidth of about 6kHz is achieved. This makes sure that the sensitivity of the load current i_{load} at low frequencies (1Hz) is also around -100dB. The resulting closed-loop Bode plot is depicted in Fig. 5. Therein, it can be seen that particularly at low frequencies, the closed loop gain and phase lag are truly flat. Besides bode plots for simulated system, Fig. 4 and Fig. 5 contain gain and phase characteristic measured on real-life PA. Notice, that these characteristics match well, which validates the amplifier model and the steps taken during the design.

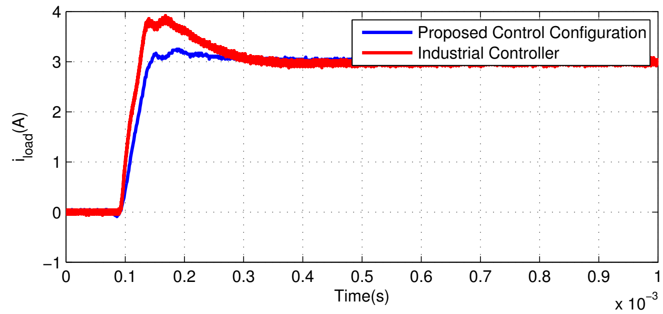


Fig. 6: Real-Time measurement with Complete Control Configuration: Step Response of $3A$.

V. Real-Time Experiments

The experimental setup used to carry out the real-time implementation and experiments is depicted in Fig. 9. The complete architecture of Fig. 2 is synthesised within the Virtex-6 FPGA on-board the power amplifier. Prototyping the control system is performed in SIMULINK with the Xilinx System Generator blockset [17]. To speed-up the deployment process, the VHDL code is generated automatically from the SIMULINK environment.

The open-loop and the closed-loop Bode plots for the system with complete control configuration are measured with Spectrum Analyser, in order to validate the simulation results. Slight difference between the simulated and measured data can be seen in Fig. 4 and Fig. 5. This can be explained by model mismatch and component variations.

The measurement results for a step reference of $3A$ and $10A$ are depicted in Fig. 6 and Fig. 7, respectively, and are compared to the measurement results of a typical industrial controller. It can be seen, that there is a significant improvement, after using the complete control configuration presented in this paper. The overshoot and settling times are reduced, while all the frequency domain specifications are met. However, the rate limiter impacts the power bandwidth of the amplifier, which could be improved by designing a more sophisticated reference governor.

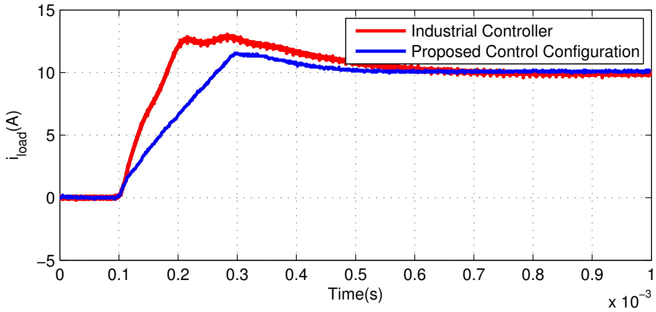


Fig. 7: Real-Time measurement with Complete Control Configuration: Step Response of $10A$.

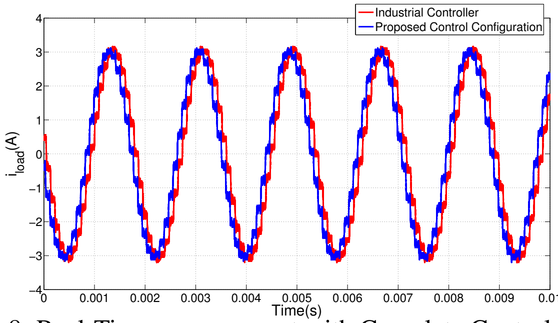


Fig. 8: Real-Time measurement with Complete Control Configuration: Sinusoidal Reference.

A varying reference is applied to the closed-system as well, i.e., the load current i_{load} for a sinusoidal reference with the amplitude of 6A peak-to-peak and a frequency of 550Hz is depicted in Fig. 8. As previously, the result is compared to that of the benchmark industrial controller. It can be seen in the figure that the proposed control configuration has smaller phase delay and no peaking as comparing to the benchmark control system.

VI. Conclusions

In this paper, a real-time control architecture was designed for the control of an industrial current amplifier. A synchronization scheme for the available measurements, along with a real-time implementation scheme were developed. An optimal controller and an observer were designed based upon the averaged discrete-time state space model of the system. An output feedback controller was designed and added to the configuration, to satisfy the frequency-domain specifications. The proposed control scheme was successfully implemented in FPGA, and real-time experiments were performed. The proposed control architecture shows consistent improvements with respect to the benchmark industrial amplifier.

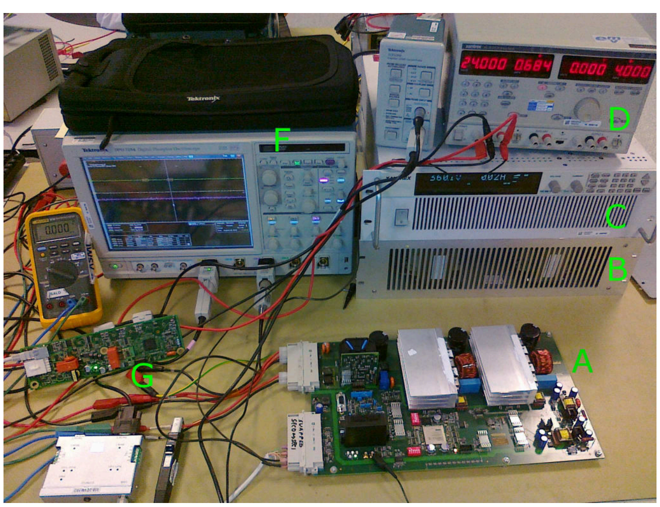


Fig. 9: Experimental Setup; (A)Current Amplifier with onboard FPGA, (B)Dummy Load, (C)High-Voltage Power Supply, (D)Low-Voltage Power Supply, (F)Oscilloscope, (G)Interface Board.

References

[1] S. Bernet, "Recent developments of high power converters for industry and traction applications," *Power Electronics, IEEE Transactions on*, vol. 15, no. 6, pp. 1102 –1117, Nov 2000.

[2] S. Kouro, M. Malinowski, K. Gopakumar, J. Pou, L. Franquelo, B. Wu, J. Rodriguez, M. Pe andrez, and J. Leon, "Recent advances and industrial applications of multilevel converters," *Industrial Electronics, IEEE Transactions on*, vol. 57, no. 8, pp. 2553 –2580, Aug. 2010.

[3] T.-S. Lee, S.-J. Chiang, and J.-M. Chang, "H_∞ loop-shaping controller designs for the single-phase UPS inverters," *Power Electronics, IEEE Transactions on*, vol. 16, no. 4, pp. 473 –481, Jul 2001.

[4] H. Takano, S. Watanabe, and M. Nakaoka, "Multiple-bridge PWM current-regulated power amplifier for magnetic resonance imaging system and its feasible digital control implementation ," in *The 25th Annual Conference of the IEEE Industrial Electronics Society*, vol. 2, 1999, pp. 785–790.

[5] S. Richter, S. Mariethoz, and M. Morari, "High-speed online MPC based on a fast gradient method applied to power converter control," in *Proceedings of the American Control Conference*, June 30 -July 2 2010, pp. 4737 –4743.

[6] S. Kouro, P. Cortes, R. Vargas, U. Ammann, and J. Rodriguez, "Model Predictive Control; A Simple and Powerful Method to Control Power Converters," *IEEE Transactions on Industrial Electronics*, vol. 56, no. 6, pp. 1826–1838, June 2009.

[7] G. Shah and S. Engell, "Tuning MPC for desired closed-loop performance for MIMO systems," in *Proceedings of the American Control Conference*, june 29 -july 1 2011, pp. 4404–4409.

[8] S. Mariéthoz and M. Morari, "Explicit model-predictive control of a PWM inverter with an *LCL* filter," *IEEE Transactions on Industrial Electronics*, vol. 56, no. 2, pp. 389–399, 2009.

[9] C. Olalla, R. Leyva, A. El Aroudi, and I. Queinnec, "Robust LQR Control for PWM Converters: An LMI Approach," *Industrial Electronics, IEEE Transactions on*, vol. 56, no. 7, pp. 2548 –2558, July 2009.

[10] H. Fukuda and M. Nakaoka, "State-vector feedback control-based 100 kHz carrier PWM power conditioning amplifier and its high-precision current-tracking scheme ," in *Proceedings of the International Conference on Industrial Electronics, Control, and Instrumentation*, Nov 1993, pp. 1105–1110.

[11] S. Watanabe, P. Boyagoda, M. Nakaoka, and H. Takano, "Analysis on a PWM power conversion amplifier with IGBT macro model to generate gradient magnetic fields in MRI systems," in *Proceedings of the IEEE International Conference on Power Electronics and Drive Systems*, vol. 1, 1999, pp. 127–132.

[12] D. G. Luenberger, "Observing the state of a linear system," *Military Electronics, IEEE Transactions on*, vol. 8, no. 2, pp. 74 –80, April 1964.

[13] I. V. Kolmanovsky, U. V. Kalabic, and E. G. Gilbert, "Developments in constrained control using reference governors," in *Proceedings of the 4th IFAC Nonlinear Model Predictive Control Conference*, Noordwijkerhout, the Netherlands, 23-27 Aug. 2012, pp. 282–290.

[14] J. Kassakian, M. Schlecht, and G. Verghese, *Principles of power electronics*, ser. Series in electrical engineering. Addison-Wesley, 1991.

[15] C. Brezinski, *Computational Aspects of Linear Control*, ser. Numerical Methods and Algorithms. Springer, 2002.

[16] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, *Feedback Control of Dynamic Systems*. Reading, MA: Addison-Wesley, 1994.

[17] Xilinx, "System generator for dsp," http://www.xilinx.com/support/sw_manuals/sysgen_ref.pdf.